Temozolomide Recurrent Therapy

INDICATIONS FOR USE:

INDICATION	ICD10	Regimen Code	HSE approved reimbursement status*
Adult patients with Grade III or IV malignant glioma, such as glioblastoma multiforme or anaplastic astrocytoma, showing recurrence or progression after standard therapy.	C71	00342a	N/A

* This applies to post 2012 indications

TREATMENT:

The starting dose of the drugs detailed below may be adjusted downward by the prescribing clinician, using their independent medical judgement, to consider each patients individual clinical circumstances.

A treatment cycle comprises 28 days. Temozolomide is administered orally once daily for the first 5 days followed by a 23 day treatment interruption (total of 28 days) until disease progression or unacceptable toxicity develops

Day	Drug	Dose	Route	Cycle (28 day)
1-5	Temozolomide	[a]200mg/m^2 ONCE daily	PO	Continuous
[a]In patients previously untreated with chemotherapy, temozolomide is administered orally at a dose of 200 mg/m^2 once daily for the first 5 days followed by a 23 day treatment interruption (total of 28 days). In patients previously treated with chemotherapy, the initial dose is 150 mg/m^2 once daily, to be increased in the second cycle to 200 mg/m^2 once daily, for 5 days if there is no haematological toxicity				
Temozolomide hard capsules should be administered in the fasting state. The capsules must be swallowed whole with a glass of water and must not be opened or chewed. If vomiting occurs after the dose is administered, a second dose should not be administered that day				
If a dose is missed, the patient should make up that dose, unless the next dose is due within 12 hours.				

ELIGIBILITY:

- Indications as above
- ECOG 0-2
- Adequate renal and hepatic function

EXCLUSIONS:

- Patients with hypersensitivity to temozolomide or any of its listed excipients
- Hypersensitivity to dacarbazine
- Severe myelosuppression
- Significant hepatic dysfunction
- Pregnancy or breastfeeding

PRESCRIPTIVE AUTHORITY:

The treatment plan must be initiated by a Consultant Medical Oncologist

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TESTS:

Baseline tests:

- FBC, renal and liver profile
- Blood glucose
- Virology screen -Hepatitis B* (HBsAg, HBcoreAb)

*(Reference Regimen Specific Complications for information on Hepatitis B reactivation)

Regular tests:

- FBC, renal and liver profile at day 1 of each cycle

Disease monitoring:

Disease monitoring should be in line with the patient's treatment plan and any other test/s as directed by the supervising Consultant.

DOSE MODIFICATIONS:

- Any dose modification should be discussed with a Consultant.
- Dose reductions and discontinuations should be applied according to Tables 1 and 2.

Table 1: Temozolomide dose levels for monotherapy treatment

Dose Level	Temozolomide Dose (mg/m^2)	Remarks
-1	100	Reduction for prior toxicity
0	150	Dose during Cycle 1
1	200	Dose during Cycles 2-6 in absence of toxicity

Table 2: Temozolomide dose reduction or discontinuation during monotherapy treatment

Toxicity	Reduce Temozolomide by 1 dose level[a]	Discontinue Temozolomide
ANC	< 1 x 10^9/L	See footnote b
Platelets	< 50 x 10^9/L	See footnote b
CTC non-haematological toxicity (except for alopecia, nausea, vomiting)	CTC Grade 3	CTC Grade 4[b]

[a]Temozolomide dose levels are listed in Table 1.

[b] Temozolomide is to be discontinued if:

- dose level -1 (100 mg/m^2) still results in unacceptable toxicity
- the same Grade 3 non-haematological toxicity (except for alopecia, nausea, vomiting) recurs after dose reduction.

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Renal and Hepatic Impairment:

Table 3: Dose modification of temozolomide in renal and hepatic impairment

Renal Impairment		Hepatic Impairment	
CrCl (mL/minute)	**Dose**	**Impairment level:**	**Dose**
≥ 36	No dose adjustment is needed	Child-Pugh A/B:	No dose adjustment is needed
< 36	No need for dose adjustment is expected	Child-Pugh C:	No need for dose adjustment is expected
Haemodialysis	No need for dose adjustment expected		
Recommendations as per Giraud et al 2023			

SUPPORTIVE CARE:

EMETOGENIC POTENTIAL:

- As outlined in NCCP Classification Document for Systemic AntiCancer Therapy (SACT) Induced Nausea and Vomiting Available on the NCCP website

Temozolomide: Moderate to high (**Refer to local policy).**

For information:

Within NCIS regimens, antiemetics have been standardised by the Medical Oncologists and Haemato-oncologists and information is available in the following documents:

- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Medical Oncology) - Available on the NCCP website
- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Haemato-oncology) - Available on the NCCP website

PREMEDICATIONS: None

OTHER SUPPORTIVE CARE:

Temozolomide can have genotoxic effects. Therefore, men being treated with it should be advised not to father a child up to 6 months after receiving the last dose and to seek advice on cryoconservation of sperm prior to treatment, because of the possibility of irreversible infertility due to therapy with temozolomide.

ADVERSE EFFECTS

- Please refer to the relevant Summary of Product Characteristics (SmPC) for details.

REGIMEN SPECIFIC COMPLICATIONS

- **Hepatitis B Virus (HBV):** Hepatitis due to HBV reactivation, in some cases resulting in death, has been reported. Experts in liver disease should be consulted before treatment is initiated in patients with positive hepatitis B serology (including those with active disease). During treatment patients should be monitored and managed appropriately.

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DRUG INTERACTIONS:

- Current SmPC and drug interaction databases should be consulted for more information.

REFERENCES:

1. Bower M, Newlands ES, Bleehan NM et al. Multicentre CRC phase II trial of temozolomide in recurrent or progressive high grade glioma. Cancer Chemother Pharmacol 1997;40:484-8.
2. Yung WKA, Prados MD, Yaya-Tur R et al. Multicenter phase II trial of temozolomide in patients with anaplastic astrocytoma or anaplastic oligoastrocytoma at first relapse. J Clin Oncol 1999;17:2762-71.
3. Giraud E L, Lijster B D, et al. Dose recommendations for anticancer drugs in patients with renal or hepatic impairment: an update. Available at: https://pubmed.ncbi.nlm.nih.gov/37269847/
4. NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting. V6 2025. Available at: https://www.hse.ie/eng/services/list/5/cancer/profinfo/chemoprotocols/nccp-classification-document-for-systemic-anti-cancer-therapy-sact-induced-nausea-and-vomiting.pdf
5. Temozolomide (Temodal ®) Summary of Product Characteristics Accessed July 2025. Available at: https://www.ema.europa.eu/en/documents/product-information/temodal-epar-product-information_en.pdf

Version	Date	Amendment	Approved By
1	20/06/2016		Prof Maccon Keane
2	20/06/2018	Updated with new NCCP regimen template and clarified treatment table	Prof Maccon Keane
3	15/07/2020	Regimen review Updated emetogenic potential	Prof Maccon Keane
4	23/07/2025	Reviewed. Updated exclusions section. Renal and Hepatic dose modifications updated to align with Giraud et al 2023. Regimen updated in line with NCCP standardisation.	Prof Maccon Keane

Comments and feedback welcome at oncologydrugs@cancercontrol.ie.

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